Design Case Study

Factors Affecting Design

Factors affecting designing + producing

Activity 1

Write down as many factors that you remember

Hint: there are 11

Factors:

- Appropriateness of design solution
- Needs
- Function
- Aesthetics
- Finance
- Ergonomics
- WHS
- Quality
- Short-term and long term environmental consequences
- Obsolescence
- Life cycle analysis

Define the factors

Activity 2

For each of these factors, write your own definition

(use the information on Google classroom to help you)

Make sure that you make note of key terms used to explain each factor

Factors:

- Appropriateness of design solution
- Needs
- Function
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- Quality
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Examples of Failed Design

Example 1

1957: The Ford Edsel

Example 2

1993: Apple Newton

Microsoft's version of the iPod

Example 3

2013: Central Park

"Sustainable Building" Broadway Sydney

Ford Edsel

Background

- 1957: car debut
- Ford had spent 10 years and $250 million on planning one of its first brand-new cars in decades.
- The Edsel came in 18 models

Ford Edsel

Background

- Henry Ford did not invent the automobile or the assembly line, he developed and manufactured the first automobile that many middle-class Americans could afford.
- His introduction of the Model T automobile revolutionized transportation and American industry.

The Edsel:

- 1957: car debut
- Ford had spent 10 years and $250 million
- The Edsel came in 18 models
- First car to be introduced by the company since 1938 when the Mercury made its debut, this twenty-year stretch more than enough time to give engineers and designers plenty to think about how to build the perfect automobile for the American consumer of the 1950s
- For many Americans, the Edsel would be the first really new car they had a chance to buy after the war, its introduction an important event in the history of automobiles.

Ford Edsel

The Edsel marketing:

- In Spring 1957, Ford launched a successful ad campaign leveraging curiosity.
- Early ads simply stated, "The Edsel is Coming", without showing the car.
- The mystery fueled public anticipation.
- Later ads revealed only glimpses—a shadow and the hood ornament.
- Anyone involved was sworn to secrecy about the Edsel.
- Dealers had to keep the car hidden or risk fines and franchise loss.
- Public anticipation peaked for the E-day launch on Sept. 4, 1957.
- Massive crowds arrived to see the car—but few actually bought it.

Ford Edsel

Branding:

- Ford positioned Edsel as a separate division, omitting the Ford name entirely. Lacking a customer base, Edsel sold just 64,000 units in its first year.

- The advertising agency involved in the rollout provided 18,000 names for Ford executives to pick from. In the end, they ignored all of these and went in their own direction.

- They named it after the first child of Ford's founder Henry and his wife Clara. However, it's just not a name that rolls off the tongue easily. When people tell their friends and neighbors what kind of car they bought, they either want name recognition or at least one that sounds cool.

Ford Edsel

Design Features:

The Edsel actually had some great innovations for its time such as a rolling dome speedometer. And its Teletouch transmission shifting system in the center of the steering wheel worked well at first.

Other design innovations kept pace with the cutting-edge accessories and trim features growing in popularity in the mid-50s. These Included ergonomically designed controls for the driver and self-adjusting brakes.

Ford Edsel

Where did it go wrong? Market Research

- The Edsel was a classic case of the wrong car for the wrong market at the wrong time
- It was also a prime example of the limitations of market research, with its “depth interviews” and “motivational” mumbo-jumbo
- On the research, Ford had an airtight case for a new medium-priced car to compete with Chrysler’s Dodge and DeSoto, General Motors’ Pontiac, Oldsmobile and Buick
- Studies showed that by 1965 half of all U.S. families would be in the $5,000-and-up bracket, would be buying more cars in the medium-priced field, which already had 60% of the market. Edsel could sell up to 400,000 cars a year.

Ford Edsel

Where did it go wrong? Market Research

- After the decision was made in 1955, Ford ran more studies to make sure the new car had precisely the right “personality.”
- Research showed that Mercury buyers were generally young and hot-rod-inclined, while Pontiac, Dodge and Buick appealed to middle-aged people.
- Edsel was to strike a happy medium. As one researcher said, it would be “the smart car for the younger executive or professional family on its way up.”
- To get this image across, Ford even went to the trouble of putting out a 60-page memo on the procedural steps in the selection of an advertising agency, turned down 19 applicants before choosing Manhattan’s Foote, Cone & Belding
- Total cost of research, design, tooling, expansion of production facilities: $250 million.

Ford Edsel

Where did it go wrong?

- The flaw in all the research was that by 1957, when Edsel appeared, the bloom was gone from the medium-priced field, and a new boom was starting in the compact car field, an area the Edsel research had overlooked completely
- In addition to the car not living up to the marketing hype, the United States was in a recession and Edsel offered its most expensive models first while other carmakers were discounting last year's models

Ford Edsel

Where did it go wrong? Manufacturing

- Ford launched the Edsel as a brand-new division, but they didn't give the car line its own manufacturing facility. Edsel relied on Ford employees to produce their cars
- Unfortunately, Ford workers resented assembling someone else's vehicle. Therefore, they took little pride in their work. Not having a separate and dedicated work force to build Edsel cars would prove to be a failure
- And for those who did buy an Edsel found that the car was plagued with shoddy workmanship. Many of the vehicles that showed up at the dealer showroom had notes attached to the steering wheel listing the parts not installed
- The Edsel's quality control issues became compounded by the Ford dealership mechanics. No additional training would lead to their unfamiliarity with the car's state-of-the-art technology. The automobiles biggest problem was its automatic "Tele-touch" transmission. The driver selected the gears by pushing buttons on the center of the steering wheel.

Ford Edsel

The End of Edsel

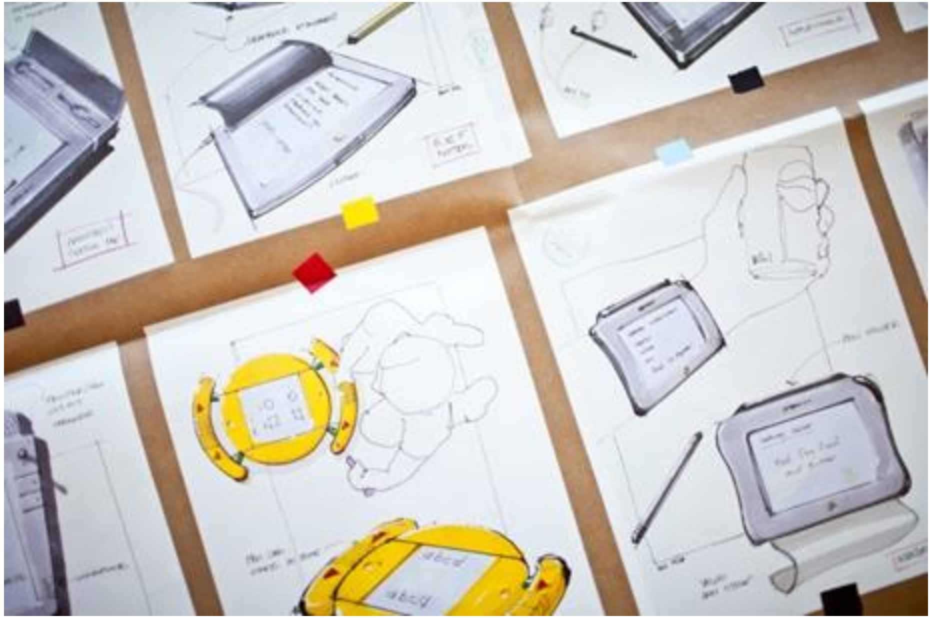
- Maybe in a different economy, with a good support system, and an honest marketing plan, the Edsel would still be around today. The company struggled for 3 years before admitting total defeat.
- When it was designed, Ford was focused on the product and not as interested in the market

Apple Newton

Background

- Newton was conceived on an airplane. That's where Michael Tchao pitched the idea to Apple's CEO, John Sculley, in early 1991.
- The first product in the Newton Line, the MessagePad 1001 went in August of 1993
- It was Apple's handheld PDA–a term Apple coined to describe it.

PDA: Personal Digital Assistant



Apple Newton

Functions

- By modern standards, it was pretty basic
- It could take notes, store contacts, and manage calendars
- You could use it to send a fax. It had a stylus, and could even translate handwriting into text
- At the time, this was highly ambitious. Handheld computers were still largely the stuff of science fiction

Design intentions:

- “The goals were to design a new category of handheld device and to build a platform to support it,” explains Steve Capps, the Newton’s head of user interface and software development who both helped dream it up and make it real. “The restrictions imposed by battery life necessitated a new architecture.” That is, with Newton, Apple didn’t just set out to create a new device. It wanted to invent an entirely new class of computing. Computers that could slip into pockets and go out into the world. In fact, the pocket was a core design requirement.

Apple Newton

Design Features

- It would have a pen, a radio that worked on a pager frequency, forms and templates built in but that could be designed on a Mac or PC, and it would have to act as a “seamless” input device for a PC
- Sculley quickly added one more core feature: size. “The number one requirement was that it had to fit in John Sculley’s pocket,” explains Gavin Ivester, who ran the Newton’s industrial design. “We focused on width because that affects how you hold it. You need to be able to reach around with your fingertips on one side and the fleshy part of your thumb on the other to feel like you’re not going to drop it. You really only feel secure if you can turn it over in your hand and still hold it.”

Apple Newton

Issues

- The result of all that work was a completely new category of device running an entirely new architecture housed in a form factor that represented a completely new and bold design language.
- There was just one problem: handwriting. “We were just way ahead of the technology,” laments Capps. “We barely got it functioning by ’93 when we started shipping it.” Handwriting recognition was supposed to be Newton’s killer feature, and yet it was the feature that probably ultimately killed the product.
- The team went back to work, and eventually got it right. But it was too late. “Character recognition got revisited and was just flawless and phenomenal,” laments Ivester. “The one stumbling block had become a joy to use, but it really never got a second look.”

Apple Newton

Issues

- Steve Jobs hated it. According to Walter Isaacson's biography of Jobs, he raged against the device for its poor performance (and because it was Sculley's innovation) and mocked its novel input mechanism.

  "God gave us ten styluses," he would say, waving his fingers. "Let's not invent another."

- And so when Jobs' at last wrestled back control of his company, he scuttled it. As he explained to Isaacson:

  "If Apple had been in a less precarious situation, I would have drilled down myself to figure out how to make it work. I didn't trust the people running it. My gut was that there was some really good technology, but it was fucked up by mismanagement. By shutting it down, I freed up some good engineers who could work on new mobile devices. And eventually we got it right when we moved on to iPhones and the iPad"

Apple Newton

Manufacturing challenges

- At the time, it was extremely difficult to get component manufacturers to build any sort of custom parts. That meant everything on the PC board had to be at right angles–which made fit a challenge. The team was trying to pull off a design referred to as “the Batman concept” a dark, sleek and sculpted aesthetic. It was also Apple’s first real departure from the Snow White design language, created by Frog Design, which had defined the company’s products in the 1980s.

Apple Newton

Legacy of the innovation

- There’s also at least one tangible thread that connects the Newton to something you likely use every day (and that indeed drives the entire mobile industry): the ARM processor. Looking to maximize battery life, the Newton team went gunning for something that could generate lots of bits MIPS per watt of power
- The best bet for that looked to be an ARM chip. Apple owned a third of the company at the time and directed development of the ARM6 processor that went into the Newton. Without the Newton, that technology could have died on the vine.

Apple Newton

Legacy of the innovation

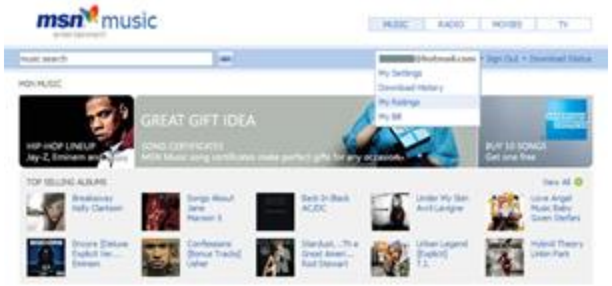
- The real impact of the Newton was the thinking that took the computer out of the office.
- Today, the PDA is with us all the time. We don't use a stylus, though, we use a keyboard (which, for most of us, is likely a much faster and more efficient way to input text.) It's our smartphone, and the whole concept of the smartphone was that it would bundle the PDA, the camera, the MP3 player, and the cell phone. And then there's Siri and Google Voice search. The idea of an intelligent assistant that can recognize natural language and act on its intent is powerful once again, but this was something Newton pioneered–one of its great strengths was its ability to take a sentence like "I have a lunch meeting with John tomorrow at noon" and translate that into an actual calendar item.

Microsoft Zune

Background

- In the early 2000s, Microsoft had the world's dominant computing platform (Windows XP)
- Microsoft had been successful in the fast changing consumer market of the 90s/2000s
  Hardware success stories in the past, including the Microsoft Mouse in the 1990s, and at the time, its Xbox gaming system

Microsoft Zune



The Zune:

- Microsoft released MSN Music in 2004 to compete with Apple's iTunes services
- In 2006, with much fanfare (including an endorsement from Barack Obama), Microsoft officially launched the Zune
- The first device was the Zune 30, released in the United States on November 2006
- It came with a capacity of 30 gigabytes, as well as a FM radio, and a 3-inch screen. The Zune 30 was initially available in black, brown or white

Microsoft Zune

The Zune:

- In terms of market share, Zune never got past the single digits (2%). The iPod on the other hand enjoyed three quarters of music-player sales worldwide
- In 2007, Microsoft launched the Zune 80, alongside the smaller Zune 4 and Zune 8 to compete with Apple's iPod nano line.
- And then in 2009 it launched the Zune HD, which was the first touch screen Zune. Whilst the Zune HD gathered good reviews, it was still a Zune to the general public, and it just didn't sell
  Zune HD promo video
  https://www.youtube.com/watch?v=iU4KCXC-M_E
- Microsoft released a slightly upgraded version in 2010, but by October 2011 Microsoft pulled the plug

Microsoft Zune

Design Features:

- Some preferred the design of the Zune HD, which was a less flashy device compared to the iPod Touch line, with a utilitarian, brushed-metal back panel that was pinned on with four prominent screws
- The Zune HD had bright OLED text-only home screen, and boasted a boldly original design compared to the iPod and iPhone's illustrated icons
- If a user tapped on a Zune menu item, the device dived into the function with a quick, animated zoom. Navigation therefore was quick and ease, and some reviewers found it to be every bit as functional as the iPod Touch's interface
- Another Zune innovation was its wireless connection. This could allow for example owners to share music with each other. But if a user was passed a copyright song by another Zune user, they could play it only three times (over three days).

Microsoft Zune

Issues:

- Microsoft was also having to contend with the much larger trend of standalone music players giving way to smartphones
- The Zune did have some problems. For example in December 2008, many first generation Zune 30 models froze. Microsoft blamed the problem on an internal clock driver written by Freescale and the way the device handled a leap year
- In order to become the dominant media player Microsoft needed a device that could make the iPod look old-fashioned. And the the Zune HD didn’t do that- there was no real reason to buy the Zune unless you were firmly in the anti-Apple camp
- Apple’s marketing power at this stage was simply too much of a hurdle for the Zune
- Officially retired in 2015

Microsoft Zune

Design Process issues

- The market needed a competitor to the iPod, it wanted one and the Zune could have been a great one—if it had been put through an effective design process
- One very important aspect of any product or service design process is the marketing and advertising. If there is a lack of clarity in the process, the marketing team will have a difficult time preparing quality campaigns. And this is exactly what happened with the Zune. The marketing team did not have a clear picture of what the device was for. They geared their campaigns toward a far-too specific audience with very artsy advertisements, and missed out on the broader audience of music listeners

Microsoft Zune

Design Process issues

- Another major design process failure that Microsoft made in designing the Zune was that they failed to bring others in. More specifically, the music industry.
- In order to make the Zune successful, Microsoft should have partnered more closely with the music industry to make them see how having competing devices and multiple music player and purchasing platforms would provide them with more leverage in the long-run. They neglected to do this, though, resulting in many of the Zune features, that could have been previously negotiated, being blocked.

Central Park

Background

- Initially built in 2013 as a residential and comercial building. It includes a mall
- five start green rating receiving the top mark for being environmentally friendly
- Urban renewal project replacing an old brewery
- Premise was to add greenery in a predominantly urban environment by adding approximately 30000 plants
- LEED Gold Certification: Recognized for sustainability and energy efficiency.

Central Park

Design Features

- Innovative Architecture: Designed by Jean Nouvel, the building has a striking, modern design with cantilevered sections.
- Mixed-Use Development: Includes residential apartments, retail spaces, offices, and dining areas.
- LEED Gold Certification: Recognized for sustainability and energy efficiency.
- Automated Irrigation System: Uses recycled water to maintain the greenery on the façade.
- Public Spaces & Art Installations: Features artwork and sculptures integrated into the development.
- Proximity to Transport & Amenities: Located near Central Station, UTS, and Broadway Shopping Centre.
- Smart Building Technology: Uses advanced lighting, energy management, and ventilation systems.

Central Park

Design Process issues

- Structural Challenges with the Heliostat: The large cantilevered heliostat required advanced engineering to ensure stability and proper light reflection.
- Green Wall Maintenance & Longevity: The vertical gardens designed by Patrick Blanc needed careful planning for irrigation, plant selection, and long-term maintenance.
- Water Recycling & Sustainability Integration: Ensuring effective integration of water recycling systems while maintaining aesthetics and functionality.
- Compliance with Local Planning Regulations: Meeting stringent Sydney city council requirements for height, shading impact, and sustainability.
- Cost & Budget Management: Balancing the high costs of advanced green technologies with commercial feasibility.
- Material Selection & Durability: Choosing materials that could withstand Sydney's climate while supporting sustainable design goals.
- Construction Complexity: Coordinating multiple contractors and specialists for the heliostat, green walls, and mixed-use spaces.
- User Experience vs. Environmental Goals: Balancing aesthetics and comfort for residents while achieving sustainability targets.
- Integration with Existing Infrastructure: Ensuring smooth connectivity with surrounding roads, public transport, and existing buildings.
- Heat & Sunlight Management: Designing the building to maximize natural light while preventing excessive heat gain inside.

Central Park

Design Process issues

- **High Maintenance Costs:** The **vertical gardens** require extensive upkeep, leading to high maintenance costs and operational challenges. Some sections have **died or needed replanting**, which questions their long-term sustainability.
- **Heliostat Ineffectiveness:** The **cantilevered heliostat** was meant to reflect light into shaded areas, but its actual impact has been debated, with **limited effectiveness and benefits** compared to initial expectations.
- **Sustainability Issues:** Despite its **green credentials**, some features (like the gardens and water systems) **consume more energy and resources than expected**, undermining its sustainability claims.
- **Building Performance vs. Design Intent:** The **solar shading elements** and **façade greenery** do not always function as intended, sometimes causing **overheating in apartments** rather than providing cooling benefits.

Central Park

Design Process issues

- **Poor Usability & Comfort:** Some residents have complained about **ventilation issues, overheating, and limited privacy** due to the building's glass-heavy and open design.
- **Aesthetic Decline Over Time:** While originally visually striking, some of the green features have **deteriorated over time**, reducing the building's iconic appeal.
- **Structural & Engineering Complications:** The **complex façade and unique cantilevered sections** required specialized engineering, increasing **construction costs and long-term maintenance burdens**.
- **Commercial Viability Issues:** Retail and public spaces **have struggled to attract consistent foot traffic**, leading to concerns about the area's economic sustainability.
- **Water Recycling Inefficiencies:** The integrated **water recycling and irrigation systems** have proven **difficult to maintain**, leading to higher operational costs than anticipated.
- **Mixed-Use Space Challenges:** Combining residential, commercial, and public spaces has led to **conflicting priorities**, with some elements not fully optimized for their intended use.

Questions

Use the terminology from the factors affecting design

- What could have been done differently?
- What have we learned from these failed design examples?


Defining the problem or initial meeting with client
Conducting a life-cycle analysis
Feedback from client or target market
Developing production schedules
Researching design ideas, general and specific criteria
Experimenting with tools, techniques and materials
Final evaluation
Finalising idea
Creating initial ideas/concept board
Revising design
Marketing
Creating prototypes
Refining ideas for the design solution
Finalising design
Production

Successful and Failed Design

Apple iPod

Homework
Describe a failed design you encountered in your daily life

What makes it unsuccessful?

- How the Apple iPod broke all Sony's Walkman Rules
- The Birth of the iPod

Read the 2 articles and answer the following questions:

Compare the success of the iPod and one of the failed design examples. What are the key factors that created a design success vs a design failure?